

HW1 Solution

Question 1.1.5

Consider $\mathbb{Z}/n\mathbb{Z}$ for some $n > 1$.

Note $\bar{0} \in \mathbb{Z}/n\mathbb{Z}$ has no multiplicative inverse. That is, there does not exist $\bar{a} \in \mathbb{Z}$ such that $\bar{0} \cdot \bar{a} = \overline{0 \cdot a} = \bar{1}$

Question 1.1.7

To check that $x \star y$ is a well-defined binary operation, we just have to check that it is a map $G \times G \rightarrow G$. The only question here is [whether the fractional part of a number is in \$G\$](#) . Although obvious, here is the proof. We use the definition of $\lfloor x + y \rfloor$:

$$\lfloor x + y \rfloor \leq x + y < \lfloor x + y \rfloor + 1$$

$$0 \leq x + y - \lfloor x + y \rfloor < 1$$

$$0 \leq x \star y < 1$$

and so $x \star y \in G$ and $\star$ is a well-defined binary operation.

We need to show that $(G, \star)$ is an abelian group. Let us first check the group axioms:

- **Associativity:** We need to check that $(a \star b) \star c = a \star (b \star c)$. We will use the general fact that for an integer k and a real number x we have

$$\lfloor x + k \rfloor = \lfloor x \rfloor + k, \quad x \in \mathbb{R}, k \in \mathbb{Z}$$

meaning that **adding an integer doesn't affect the floor function** (or any rounding).

We now calculate $a \star b \star c$ in the two possible ways:

$$\begin{aligned} (a \star b) \star c &= (a + b - \lfloor a + b \rfloor) \star c \\ &= (a + b - \lfloor a + b \rfloor) + c - \lfloor (a + b - \underbrace{\lfloor a + b \rfloor}_{\in \mathbb{Z}}) + c \rfloor \\ &= a + b + c - \lfloor a + b + c \rfloor \\ a \star (b \star c) &= a \star (b + c - \lfloor b + c \rfloor) \\ &= a + (b + c - \lfloor b + c \rfloor) - \lfloor a + (b + c - \underbrace{\lfloor b + c \rfloor}_{\in \mathbb{Z}}) \rfloor \\ &= a + b + c - \lfloor a + b + c \rfloor \end{aligned}$$

We see that in both cases we get the same answer, thereby proving the associative property.

- **Identity:** We need to find an element which leaves every element of the group unchanged. This is easily seen to be 0. Indeed, for any $x \in G$:

$$\begin{aligned} 0 \star x &= 0 + x - \lfloor 0 + x \rfloor = x \\ x \star 0 &= x + 0 - \lfloor 0 + x \rfloor = x \end{aligned}$$

Notice that for $x \in G$, $\lfloor x \rfloor = 0$ because we're just rounding down a number from $[0, 1)$.

- **Inverse:** We need to check if every element has an inverse with respect to the above operation. We already know that 0 has an inverse (itself) and for every other element, $1 - x$ will be their inverse. Let $x \in (0, 1)$, then:

$$\begin{aligned} x \star (1 - x) &= x + 1 - x - \lfloor x + 1 - x \rfloor = 1 - 1 = 0 \\ (1 - x) \star x &= 1 - x + x - \lfloor 1 - x + x \rfloor = 1 - 1 = 0 \end{aligned}$$

showing that every element has an inverse.

This shows that $(G, \star)$ is a group.

Finally, let us show that this is an [abelian](#) group. A group is abelian (or commutative) if $x \star y = y \star x$. We check that this is true. Indeed

$$\begin{aligned} x \star y &= x + y - \lfloor x + y \rfloor \\ &= y + x - \lfloor y + x \rfloor \\ &= y \star x \end{aligned}$$

Question 1.1.20

First suppose $|x| = n < \infty$. Then $x^n = e$. Multiply both sides by $(x^n)^{-1}$ to get $e = (x^n)^{-1}$. By exercise 19, we know that $(x^n)^{-1} = x^{-n} = (x^{-1})^n$. Hence $(x^{-1})^n = e$. Thus $|x^{-1}| \leq n = |x|$. A very similar argument shows $|x| \leq |x^{-1}|$. Hence $|x| = |x^{-1}|$.

Now suppose that x has infinite order. Then there does not exist an $n \in \mathbb{Z}^+$ such that $x^n = e$. But suppose that x^{-1} has finite order. Then, by the unstated "similar argument" from the first panel, $|x| \geq |x^{-1}| < \infty$. This contradiction shows that x^{-1} has infinite order as well. By reversing the roles of x and x^{-1} in this panel, we also see that x^{-1} has infinite order $\implies x$ has infinite order.

This exhausts all possibilities and hence x and x^{-1} always have the same order.

Question 1.1.29

$\implies$: First suppose $A \times B$ is abelian. Then, for any $(a_1, b_1), (a_2, b_2) \in A \times B$,

$$(a_1 a_2, b_1 b_2) = (a_1, b_1)(a_2, b_2) = (a_2, b_2)(a_1, b_1) = (a_2 a_1, b_2 b_1)$$

From the definition of the Cartesian product, we know $(a_1 a_2, b_1 b_2) = (a_2 a_1, b_2 b_1)$ if and only if $a_1 a_2 = a_2 a_1$ and $b_1 b_2 = b_2 b_1$. As a_1, a_2 are arbitrary elements of A and b_1, b_2 are arbitrary elements of B , we conclude that A and B are abelian.

$\Leftarrow$: Now suppose A and B are abelian. Then, for any $a_1, a_2 \in A$ and $b_1, b_2 \in B$,

$$(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2) = (a_2 a_1, b_2 b_1) = (a_2, b_2)(a_1, b_1)$$

As (a_1, b_1) and (a_2, b_2) are arbitrary elements of $A \times B$, we conclude that $A \times B$ is abelian.

Question 1.1.31

Let $t(G)$ denote the set $\{g \in G \mid g \neq g^{-1}\}$ and let $g \in t(G)$. Then, because $g \neq g^{-1} \implies g^{-1} \neq (g^{-1})^{-1} (= g)$, we see that g^{-1} is also in $t(G)$. Moreover, g and g^{-1} are distinct elements in $t(G)$. Because we can pair up every element of $t(G)$ with its distinct inverse, we see that the order of $t(G)$ is even.

As $|G|$ and $|t(G)|$ are even, $|G - t(G)|$ must also be even ($|G| = |t(G)| + |G - t(G)|$). But we have to prove that it contains at least one element that has order 2. Note that $e \notin t(G)$ because $e = e^{-1}$. So $e \in G - t(G)$. Because $|G - t(G)|$ is even there must be at least one other element in it. But every other $g \in G$ such that $g = g^{-1}$ has order two ($g = g^{-1} \implies g^2 = e \implies |g| = 1$ or $|g| = 2 \implies g = e$ or $|g| = 2$).

So there's at least one element of G with order 2.

Question 1.2.3

Suppose that an element of D_{2n} is not a power of R . Then it must be of the form sr^k for some integer k . Then

$$(sr^k)^2 = (sr^k)(sr^k) = s(sr^{-k})r^k = s^2 r^{k-k} = 1.$$

So the element must at least be of order 2. We know that an element is of order 1 if and only if it is the group identity, and $sr^k \neq 1$ since, for instance, it should at least serve as a right-identity for s if it were the group identity, but

$$s(sr^k) = s^2 r^k = r^k \neq s.$$

So we can conclude that every element that is not a power of r is of order 2.

Now we must show that D_{2n} is generated by s and sr . Since we know that D_{2n} is generated by r and s , it suffices to show that we can generate r with s and sr . Indeed $s(sr) = r$, and so $\langle r, sr \rangle$ generates D_{2n} as well.

Question 1.2.16

Note that $D_4 = D_{2,2} = \langle r, s \mid r^2 = s^2 = 1, rs = sr^{-1} \rangle$. So making the identification $x_1 = r$ and $y_1 = s$, we immediately see that r and s satisfy $x_1^2 = y_1^2 = 1$. As for $(x_1 y_1)^2 = 1$, note that

$$(x_1 y_1)^2 = (rs)^2 = rsrs = rr^{-1}ss = 1$$

and

$$rs = x_1 y_1 = (x_1 y_1)^{-1} = y_1^{-1} x_1^{-1} = y_1 x_1^{-1} = sr^{-1}$$

Thus r, s satisfy the relations on x_1 and y_1 and x_1 and y_1 satisfy the relations for r and s . Hence

$$D_4 = \langle r, s \mid r^2 = s^2 = 1, rs = sr^{-1} \rangle = \langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$$

Question 1.1.33

First a lemma: If $|a| = n < \infty$ and $a^m = e$, then n divides m (i.e. $m = nk$ for some integer k).

Proof: By the division algorithm, there exist integers q, r such that $m = nq + r$ where $0 \leq r < n$. Hence

$$\begin{aligned} a^m &= a^{nq+r} \\ &= a^{nq} a^r \\ &= (a^n)^q a^r \\ &= e^q a^r \\ &= a^r \end{aligned}$$

But this implies $a^r = 1$. If $r > 0$, then this contradicts the fact that a has order n (because $r < n$). Hence $r = 0$. Thus $m = nq$ for some $q \in \mathbb{Z}$. Thus n divides m , as desired.

(a) Let $x \in G$ such that $|x| = n < \infty$ and n is odd. Now suppose there exists an $i \in \{1, 2, \dots, n-1\}$ such that $x^i = x^{-i}$ and we will proceed to prove a contradiction. $x^i = x^{-i}$ implies $e = (x^i)^2 = x^{2i}$. From the above lemma, this implies that n divides $2i$. Notice that $0 < i < n$ implies $0 < 2i < 2n$. From this and the fact that n divides $2i$ we conclude that n must equal $2i$. But $2i$ is even and n is odd. Contradiction. Hence there is no $i \in \{1, 2, \dots, n-1\}$ such that $x^i = x^{-i}$.

(b) Let $x \in G$ such that $|x| = n = 2k < \infty$ where $k \in \mathbb{Z}^+$.

$\implies$: First suppose there is some $i \in \{n-1\}$ such that $x^i = x^{-i}$. Using the same argument as in part (a) (but this time we don't get a contradiction because n is even), we see that $2k = n = 2i$. Hence $k = i$.

$\impliedby$: Now consider the following

$$(x^k)^2 = x^{2k} = x^n = e$$

This implies that $x^k = x^{-k}$. I.e. $i = k \implies x^i = x^{-i}$.

Thus $x^i = x^{-i} \iff i = k$.